Aditya singh

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CISCO PACKET TRACER INTERNSHIP PROJECT

Design & implementation of a coverage data & voice enterprise

1. INTRODUCTION

NAME: ADITYA SINGH
ROLL NO : 25SCS1003001266
COLLAGE NAME: IILM UNIVERSITY GREATER NOIDA

2. Project Aim

The aim of this project is to design and implement a converged enterprise network that supports both data and voice communication using Cisco Packet Tracer. The design combines wired and wireless users, IP phones, printers, servers, VLANs, routing devices, wireless access points, and centralized network management.

3. Project Objectives

The main objectives are to:

- Design a reliable enterprise network.
- Integrate data and voice services over an IP infrastructure.
- Separate departments using VLANs.
- Provide wired and wireless connectivity.
- Support IP telephony, servers, and shared printers.
- Enable communication between required network segments.
- Provide Internet/external connectivity.
- Provide centralized wireless and network management.
- Test the network and verify connectivity.
- Allow future expansion of users and services.

4. Network Requirements

The enterprise requires reliable data communication between departments, centralized access to servers and printers, IP-based voice communication, and wireless access for mobile users. Departmental separation is required for better organization and security, while routing is needed for communication between different IP networks. The design also requires external connectivity and manageable network infrastructure.

5. Network Architecture and Topology

The proposed topology represents a multi-department enterprise network. It contains Sales, Administration, Engineering, Management/server resources, wireless access points, IP phones, printers, servers, switches, routers, a Wireless LAN Controller, a Network Controller, and an external network connection.

The central switching and routing infrastructure connects the departmental networks with the server farm, wireless infrastructure, management devices, and external network. This centralized architecture provides a structured and scalable network design.

Figure 1: Overall Enterprise Network Topology

Insert the Packet Tracer topology screenshot here.

6. VLAN and IP Addressing Design

VLANs divide the physical network into logical networks. This reduces broadcast traffic and provides departmental separation. The topology shows the following VLANs:

VLAN	Department/Network	Purpose
VLAN 15	Sales	Sales users and devices
VLAN 20	Administration	Administrative users and devices
VLAN 30	Engineering	Engineering users and devices
VLAN 50	Management/Server-related network	Management and server resources

The diagram shows 192.168.2.1 for the Administration network and 192.168.3.1 for Engineering. These values should be verified against the actual Packet Tracer configuration before final submission. The complete IP addressing table should contain the network address, subnet mask, gateway, and assignment method for every VLAN.

Network	Gateway	Subnet Mask	Assignment
Sales VLAN	[Verify]	[Verify]	Static/DHCP
Administration VLAN	192.168.2.1*	[Verify]	Static/DHCP
Engineering VLAN	192.168.3.1*	[Verify]	Static/DHCP
Management/Server VLAN	[Verify]	[Verify]	Static/DHCP

7. Network Devices and Services

The network uses switches to connect computers, IP phones, printers, and access points. The multilayer switching infrastructure can provide Layer 3 routing and default gateways for VLANs. Routers connect different networks and provide external connectivity.

Wireless access points provide mobile connectivity, while the Wireless LAN Controller provides centralized wireless management. Cisco IP phones provide voice communication over the IP network. Printers allow departmental resource sharing, and the server farm provides centralized network and application services.

The exact services running on each server should be taken from the Packet Tracer configuration. Possible services include DHCP, DNS, web, file transfer, and email services.

8. Wireless and Voice Network Design

Wireless access points are distributed across the enterprise areas to provide coverage for laptops and other wireless clients. Centralized management through the Wireless LAN Controller simplifies configuration and monitoring. Appropriate authentication and encryption should be used to protect wireless access.

Voice is an important part of the converged design. Cisco IP phones are deployed in the departmental areas, allowing voice traffic to use the same IP infrastructure as data traffic. A dedicated voice VLAN can be used if configured. Quality of Service should also be considered because voice is sensitive to delay, jitter, and packet loss.

9. Routing, Network Services, and Internet Connectivity

Because each VLAN represents a separate IP network, Layer 3 routing is required for communication between VLANs. The general path is:

End Device → Switch → Layer 3 Gateway → Destination Network → Destination Device

Inter-VLAN routing allows authorized users to reach servers and users in other departments. Routing policies can also be used to control access to sensitive management or server networks.

The enterprise is connected to an external network through a router and cable modem/cloud infrastructure. If NAT is configured, internal private addresses can be translated for external communication.

Centralized services may include DHCP for address assignment, DNS for name resolution, web/file/email services, printing, and network management. Only services actually configured should be reported as implemented.

10. Network Security and Management

Security is provided through logical segmentation and should be strengthened using appropriate controls. VLANs separate departments, while ACLs can restrict communication between sensitive networks. Switch port security can prevent unauthorized devices from using selected ports, and SSH should be preferred for secure device administration. Unused switch ports should be disabled where appropriate. Wireless networks should use secure authentication and encryption.

The management network and Network Controller provide a basis for monitoring and administering the infrastructure. Separating management traffic from normal user traffic improves organization and can reduce unauthorized access.

11. Implementation in Cisco Packet Tracer

The network was implemented by placing the required routers, switches, servers, computers, IP phones, printers, access points, and controllers in Cisco Packet Tracer. The devices were connected according to the topology, VLANs were created and assigned, IP addresses were configured, and routing was established between the required networks. Wireless components and IP phones were then configured according to the design, followed by server and network-service configuration.

The final implementation should be supported with screenshots showing important configurations.

12. Testing and Verification

Testing is required to confirm that the network operates correctly. The following tests should be performed:

Test	Source	Destination	Expected Result
1	PC	Default gateway	Successful
2	PC	Same-VLAN device	Successful
3	Sales PC	Engineering PC	Successful if permitted
4	Engineering PC	Server	Successful
5	Wireless laptop	Gateway	Successful
6	IP phone	Voice destination	Successful if configured
7	Internal PC	External server	Successful if configured

Useful Cisco verification commands include `show vlan brief`, `show interfaces trunk`, `show ip interface brief`, `show ip route`, `show running-config`, and `show mac address-table`. Connectivity can be checked using `ping` and `traceroute`.

13. Troubleshooting

Common problems can result from incorrect IP addresses, VLAN assignments, trunk configuration, routing, wireless settings, or voice configuration. Troubleshooting should begin with physical connectivity, then Layer 2 configuration, Layer 3 addressing and routing, and finally application services.

Problem	Possible Cause	Solution
No gateway connectivity	Incorrect IP/gateway	Verify addressing
VLAN communication failure	Wrong VLAN assignment	Check VLAN configuration
Inter-VLAN failure	Routing problem	Check Layer 3 interfaces/routes
Wireless failure	AP/WLAN configuration	Verify wireless settings
Phone failure	Voice configuration	Check phone and switch settings
Server unreachable	Addressing/routing issue	Verify server and routes

14. Performance, Scalability, and Advantages

The proposed design is scalable because additional users can be added to existing VLANs, while new departments can receive separate VLANs. Additional switches, access points, servers, and IP phones can also be introduced as the organization grows.

The main advantages are the integration of data and voice, logical network segmentation, centralized wireless and network management, wireless mobility, centralized services, resource sharing, and easier troubleshooting.

15. Limitations and Future Improvements

Cisco Packet Tracer is a simulation environment and cannot fully reproduce the conditions of a physical enterprise network. Real-world factors such as hardware failures, wireless interference, physical cabling problems, traffic load, and actual voice quality are not completely represented.

Future improvements could include enterprise firewalls, stronger ACL policies, Quality of Service for VoIP, redundant switches and routers, backup links, advanced wireless security, intrusion detection, centralized logging, SNMP monitoring, IPv6, and improved high availability.

16. Results and Discussion

The proposed design demonstrates how a single enterprise IP infrastructure can support departmental data networks, wireless access, IP telephony, centralized servers, and external connectivity. VLANs provide logical separation, while routing enables communication between authorized networks. Wireless access points provide mobility and the Wireless LAN Controller supports centralized wireless management.

The final results should be supported by actual Packet Tracer screenshots and test outputs. Successful gateway, inter-VLAN, server, wireless, voice, and external connectivity tests will demonstrate whether the design meets its objectives.

17. Conclusion

The project demonstrates the design and implementation of a converged enterprise network supporting both data and voice communication. The network combines VLAN segmentation, IP addressing, routing, wireless access, IP telephony, servers, printers, network management, and external connectivity.

The use of VLANs improves organization and logical separation, while Layer 3 routing provides communication between required networks. Wireless infrastructure provides mobility, and IP phones demonstrate the integration of voice into the same IP environment. Cisco Packet Tracer provides a practical platform for implementing and testing the proposed design before real-world deployment.

Overall, the project demonstrates the importance of structured network planning, addressing, segmentation, routing, security, management, and testing in building a reliable and scalable enterprise network.

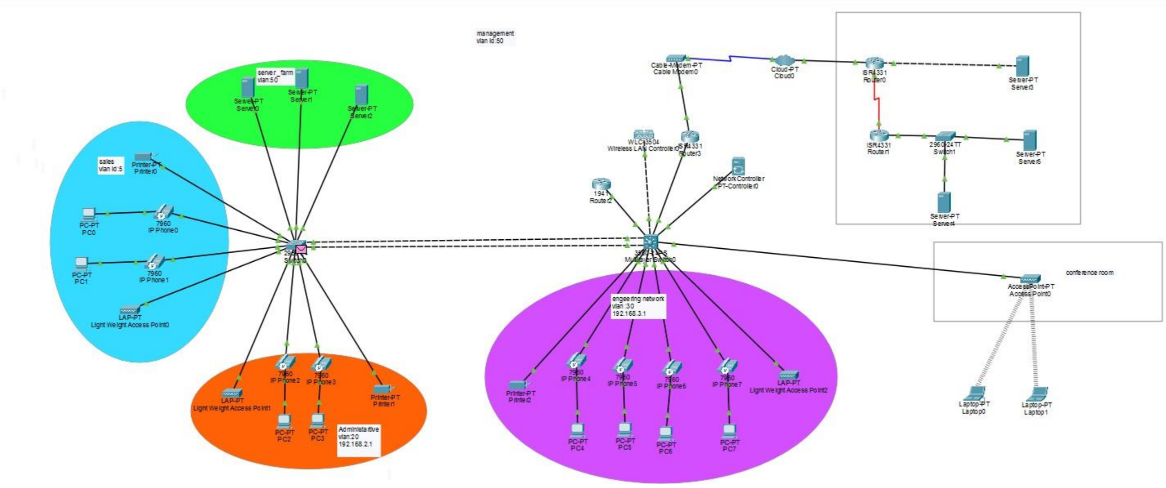
18. References

References should include Cisco networking and Packet Tracer documentation, Cisco IOS documentation, relevant networking textbooks, RFCs, course materials, and official documentation related to VLANs, IPv4, routing, VoIP, and wireless networking. The referencing style should follow the institution's required format.

19. Appendices

Include only important supporting evidence:

- Full network topology.
- VLAN configuration (`show vlan brief`).
- IP interface configuration (`show ip interface brief`).
- Routing table (`show ip route`).
- Relevant running configuration.
- Ping/traceroute results.
- Wireless configuration screenshots.
- IP phone/voice configuration screenshots.



Executive Summary

The purpose of this technical report is to document the design, logical implementation, and deployment strategies of a high-availability, converged enterprise campus network architecture modeled within Cisco Packet Tracer. Modern business operations necessitate a highly integrated network fabric capable of seamlessly hosting heterogeneous traffic classes, including real-time Voice over IP (VoIP) streams, mission-critical server pools, corporate wireless access points, and standard end-user computing terminals. This topology employs a hierarchical design model utilizing a centralized multilayer core routing fabric to interconnect diverse logical subnets. By implementing Virtual Local Area Networks (VLANs), centralized Wireless LAN Controllers (WLCs), and specialized Quality of Service (QoS) boundaries, this infrastructure mitigates broadcast domains, enhances perimeter and internal security, and establishes deterministic scalability pathways necessary for the modern enterprise ecosystem.